Computer Science & IT

Theory of Computation



Regular Languages

Lecture No. 01



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Topics to be Covered



- Topic
- Topic
- Topic

Ple

RBR

III - 86

IV - 13

ME, IISc

avg - 10

TOC



TOC - very easy
DM — easy

Basic → advanced

Problem - 2 time

Symbol — a, b, c, ..., 0, 1, ..., A, B, ...,





⇓

Alphabet Σ. eg = $\Sigma = \{a, b\}$ $\Sigma = \{0, 1\}$, $\Sigma =$ 

finite

$\Sigma = \{a, b\}$

String ⇒ any sequence of symbols from alphabet

Ex: a, b, aa, ab, aaa ....

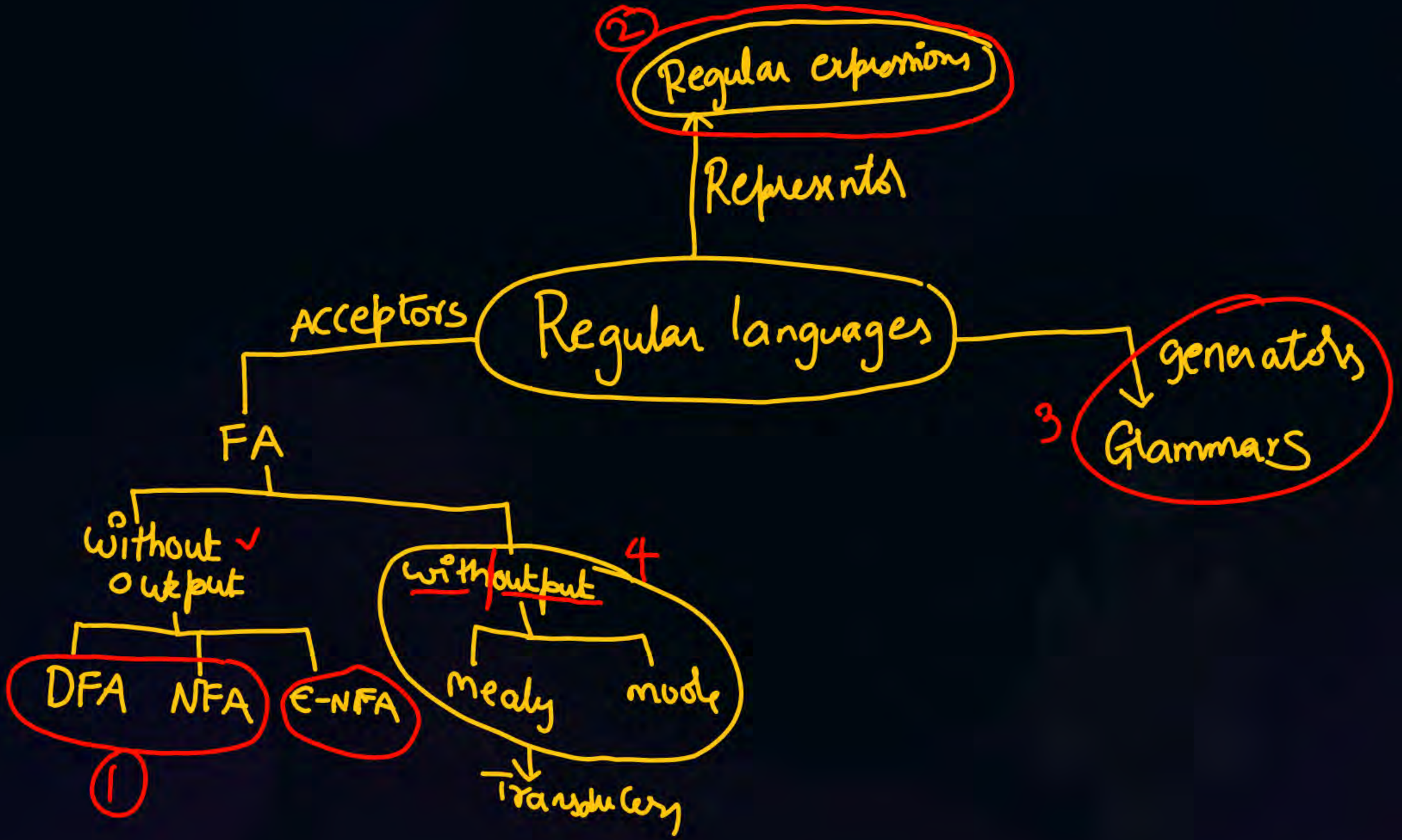
Languge ⇒ Set of strings (finite & infinite)

$\Sigma = \{a, b\}$

$L_1 = \{aa, ab, ba\} \Rightarrow$ finite

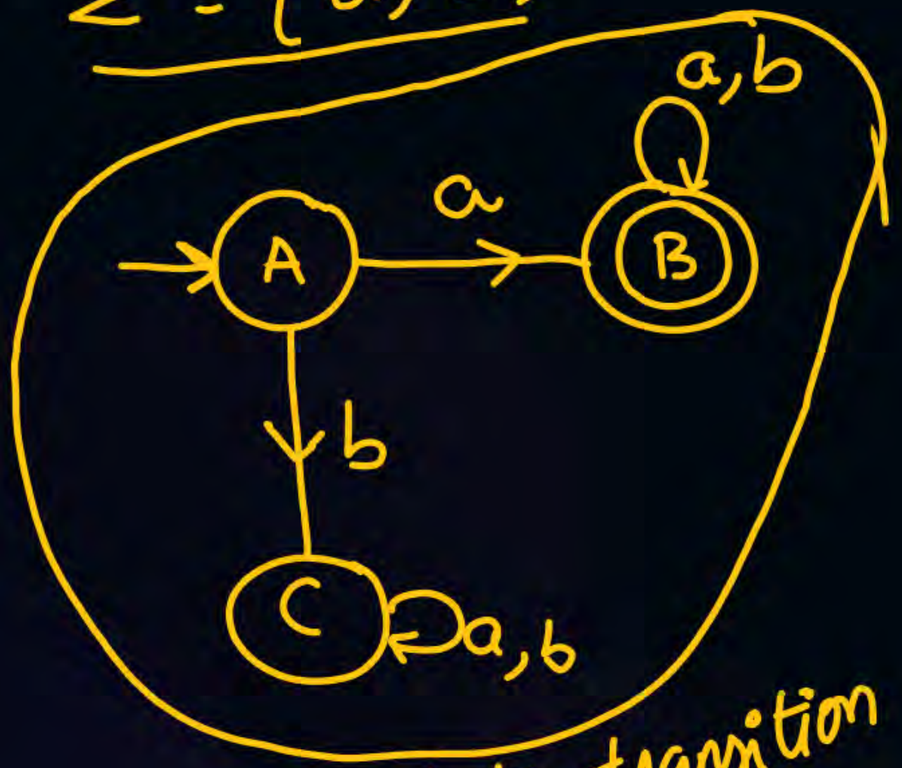
L_2 = Set of all strings starting with a

✓ $= \{a, aa, ab, aaa, aab \cdots\cdots\}$


2
Regular expressions
Represents
Regular languages
acceptors
generators
3
Grammars
FA
without output
with output
4
DFA
NFA
ε-NFA
1
Mealy
moore
Transducers

Deterministic finite automata:

$\Sigma = \{a, b\}$

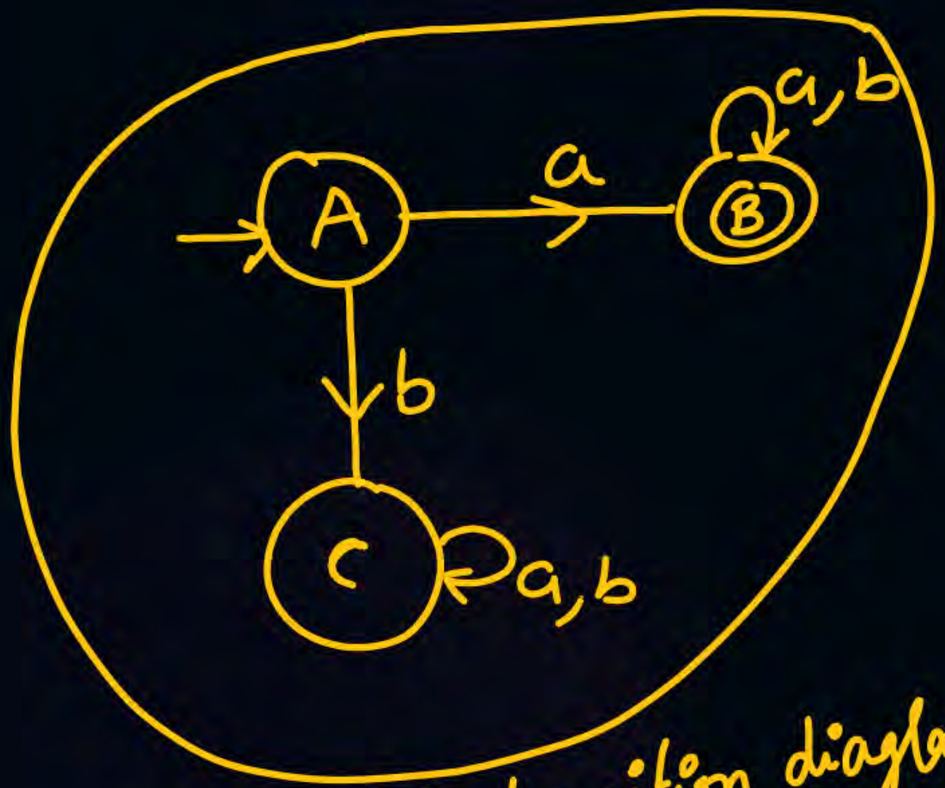


State transition diagram

States = {A, B, C}

Final state = {B}

initial state = {A}



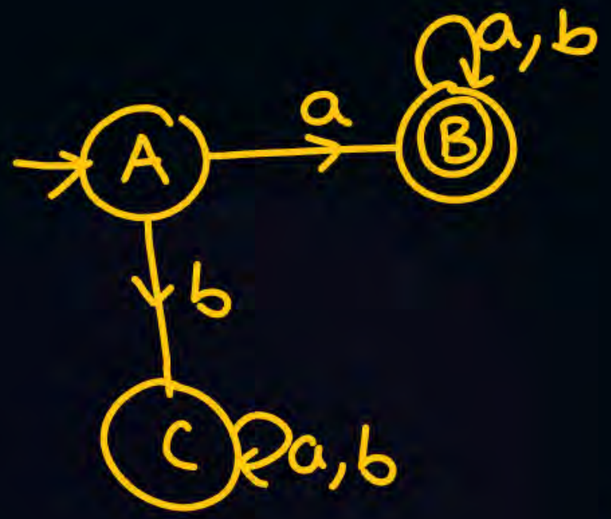
State transition diagram

	a	b
→ A	B	C
*B	B	B
C	C	C

State transition table

Definition of DFA: $Q, \Sigma, \delta, q_0, F$

- Q — Set of States
- Σ — i/p alphabet
- δ — Transition function
- q_0 — Initial State
- F — Set of final States

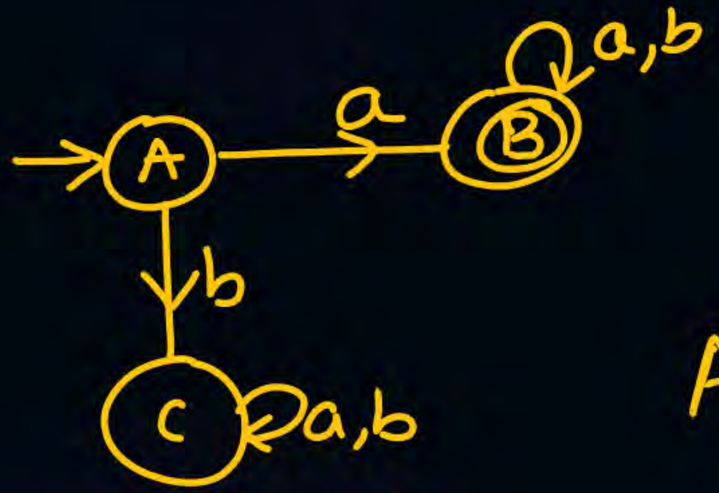


$Q = \{A, B, C\}$

$\Sigma = \{a, b\}$

$q_0 = A$

$F = \{B\}$

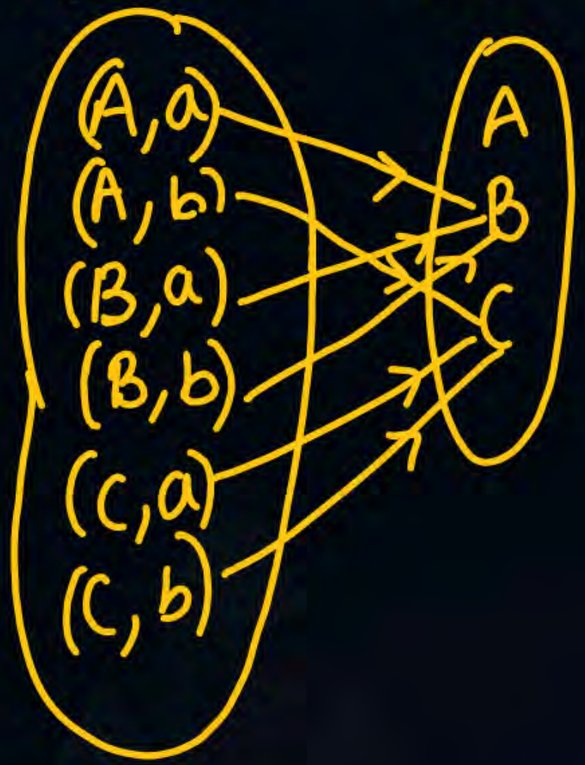


$Q = \{A, B, C\}$ (3)

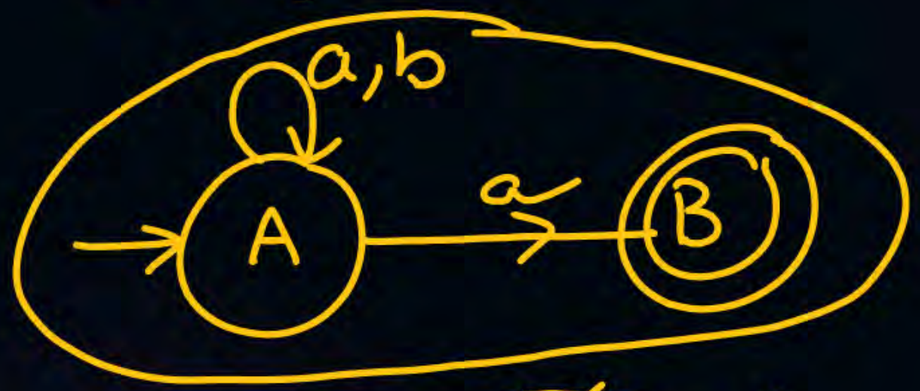
$\Sigma = \{a, b\}$ (2)

$A, a \rightarrow B$

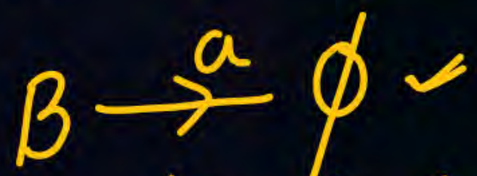
$\delta : (Q \times \Sigma) \rightarrow Q$

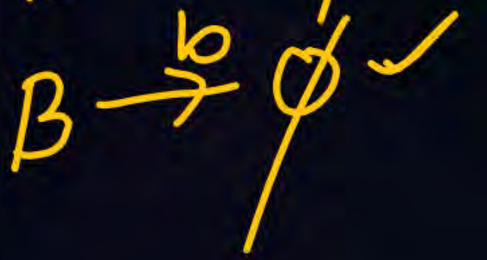


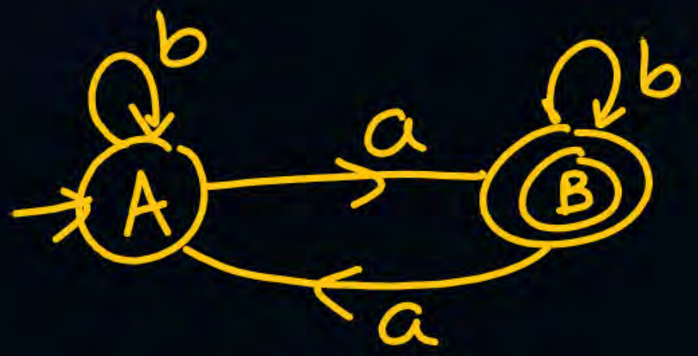
$\Sigma = \{a, b\}$





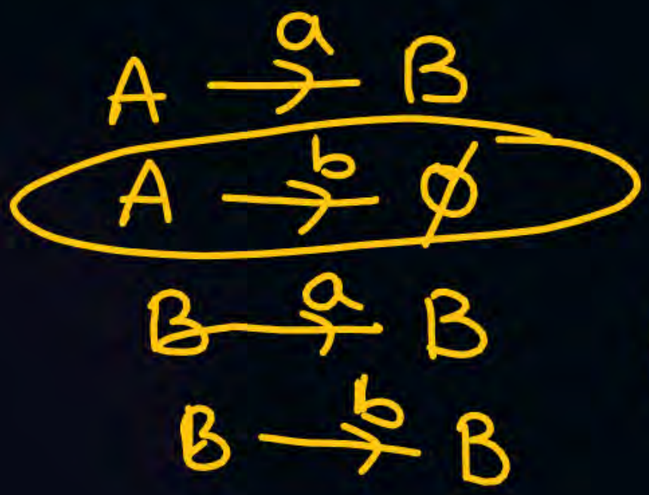






A	a	B
A	b	A
B	a	A
B	b	B



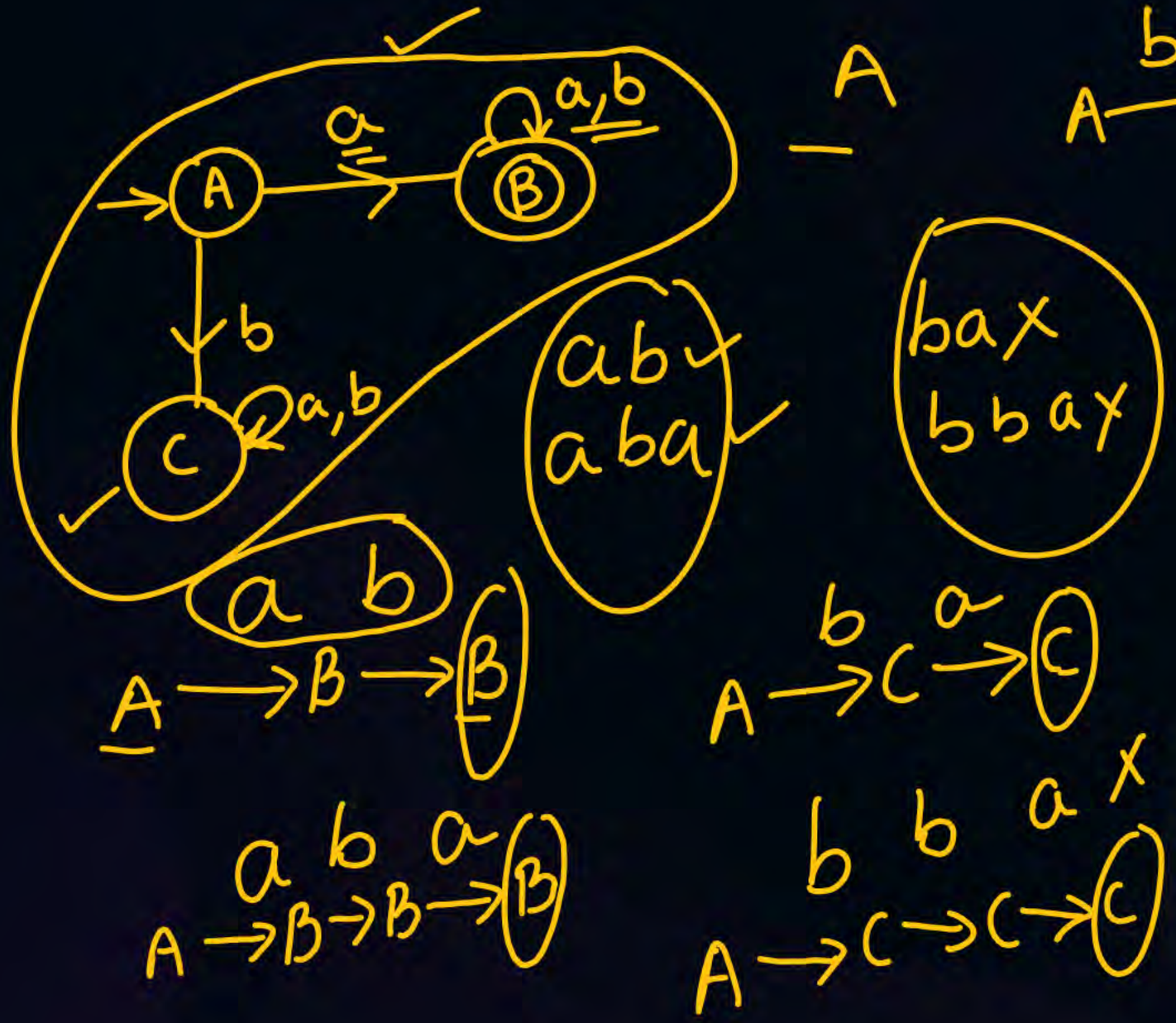


X DFA

NFA

Later


String
FA
Yes, it is in the language
No, it is not in the language



A

A → b → C ✓

L = { Set of all strings starting with a }

NFA: $(Q, \Sigma, \delta, q_0, F)$

- Q: Set of States
- Σ: i/p alphabet
- δ: Transition function
- q_0: Initial State
- F: Set of Final States

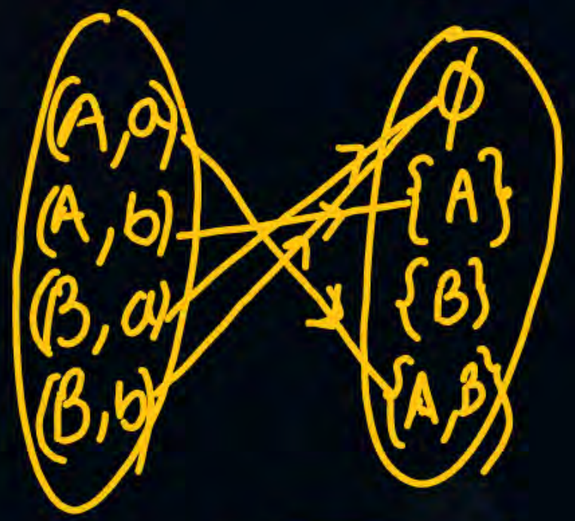


$\delta : Q \times \Sigma \rightarrow 2^Q$

$Q = \{A, B\}$

$\Sigma = \{a, b\}$

$2^Q = \{\phi, \{A\}, \{B\}, \{A,B\}\}$



$2^Q \rightarrow$ Power set $\rightarrow$ Set of all Subsets of Q

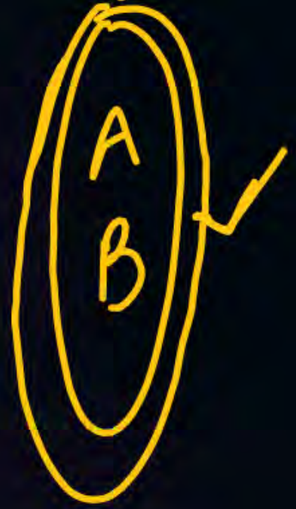
NFA

- 1 State ✓
- many State ✓
- No State ✓

$Q = \{A, B\}$

DFA

$\delta : Q \times \Sigma \rightarrow Q$

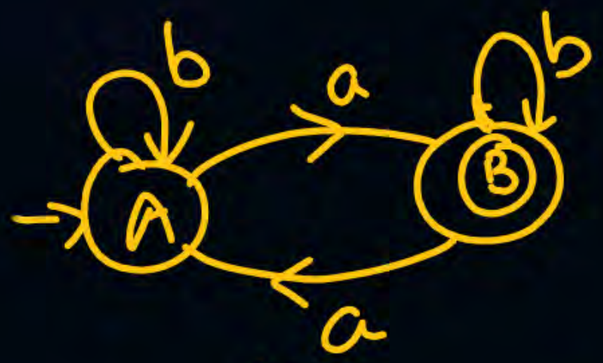


NFA

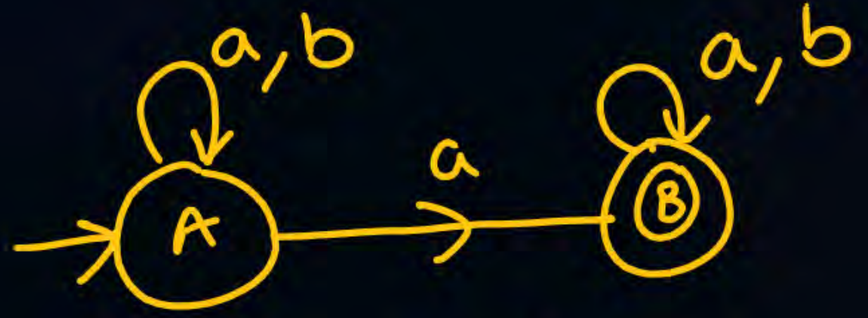
$\delta : Q \times \Sigma \rightarrow 2^Q$



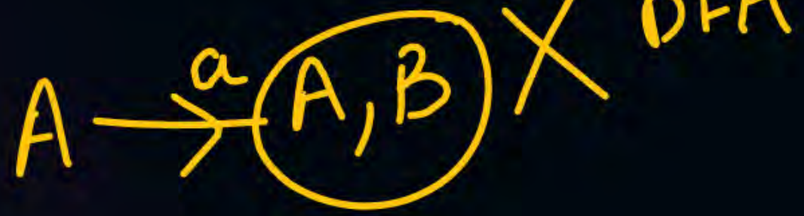




NFA & DFA



NFA



X DFA

$\Sigma = \{a, b\}$ $\quad\quad \Sigma^0$

$\Sigma^1 = \{a, b\}$ ✓

$\Sigma^2 = \Sigma \cdot \Sigma = \{a, b\} \cdot \{a, b\}$

$= \{aa, ab, ba, bb\}$ ✓

$ab \neq ba$

Σ^3 ✓ → Set of all strings of length 3 ✓

Σ^n → " " " n

$\Sigma = \{a, b\}$ $\Sigma^0 \rightarrow$ string of length 0.

$\{\in\}$

↓

epsilon $|\in| = 0$

↓

zero length string

Σ^* → Kleene closure

$$\Sigma^* = \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \Sigma^3 \ldots\ldots \Sigma^n \ldots\ldots$$

Set of all strings over Σ

= universal set



any language is a subset of Σ^*

Reg expressions :-

$\phi = \{\ \}$

$\epsilon = \{\epsilon\}$

$\phi \neq \epsilon$ ✓

operators : $+$, $\cdot$, $*$

$+$ → union, $\cdot$ → Concatination, $*$ → Kleene closure

$\phi, \epsilon, a \in \Sigma$ → primite Reg expression

$r_1 \to RE$, $r_2 \to RE$

$r_1 + r_2$, $r_1 \cdot r_2$, r_1^* → RE

$\Sigma = \{a, b\}$

'a' in a RE $\rightarrow \{a\}$

$b \rightarrow \{b\}$

$a+b \rightarrow \{a, b\}$

$a.b \rightarrow \{ab\}$

$\Sigma = \{a, b\}$

RE	language
a	$\{a\}$
b	$\{b\}$
ϵ	$\{\epsilon\}$
ab	$\{ab\}$
$a+b$	$\{a, b\}$
a^*	$\{a^0, a^1, a^2, \cdots\} = \{\epsilon, a, aa, aaa, \cdots\}$

$$(a+b)^* = (a+b)^0 \cup (a+b)^1 \cup (a+b)^2 \cup \ldots\ldots\ldots$$

$$= \epsilon \cup \{a, b\} \cup \{aa, ab, ba, bb\} \ldots\ldots$$

$\Sigma = \{a, b\}$ = universal language

b^*a

$\{\epsilon, b, bb, bbb \ldots\}.a$

$\{a, ba, bba, bbba \ldots\}$

$\epsilon.a = a$
$a.\epsilon = a$
$\epsilon.a.\epsilon = a$

$b^*a \overset{?}{=}$ ending aX

aa

$b^*a \rightarrow$ any no of b's followed by a 'a'

ab^* ✓ aa

$b^{*^2}ab^{*^1}$ ⇒ Set of all string with exactly one a

abb $bbab$ $babb$ $b^{*^1}ab^{*^2}$

$b^{*^2}ab^{*^1}$

$bbab$

$\epsilon abb = abb$

$*$ → 0, 1, 2, 3, 4, ... ✓ ✓ ✓ ✓ ✓

b^* b^*

$b^*(b^*ab^*)b^*$

$b^*.b^*$

$\{\epsilon, b, bb \ldots\}.b^*$

b^*

b^* U

$aa^* = a^*a = a^+$ → positive closure

$a.\{\epsilon, a, aa, aaa \cdots\}$

$\{a, aa, aaa, \cdots\} = a^+$ 1, 2, 3, 4, ...

a^* 0, 1, 2, 3 ....

THANK - YOU